ECO 331

Economic History

`Automation, Technology, and Labor Demand`

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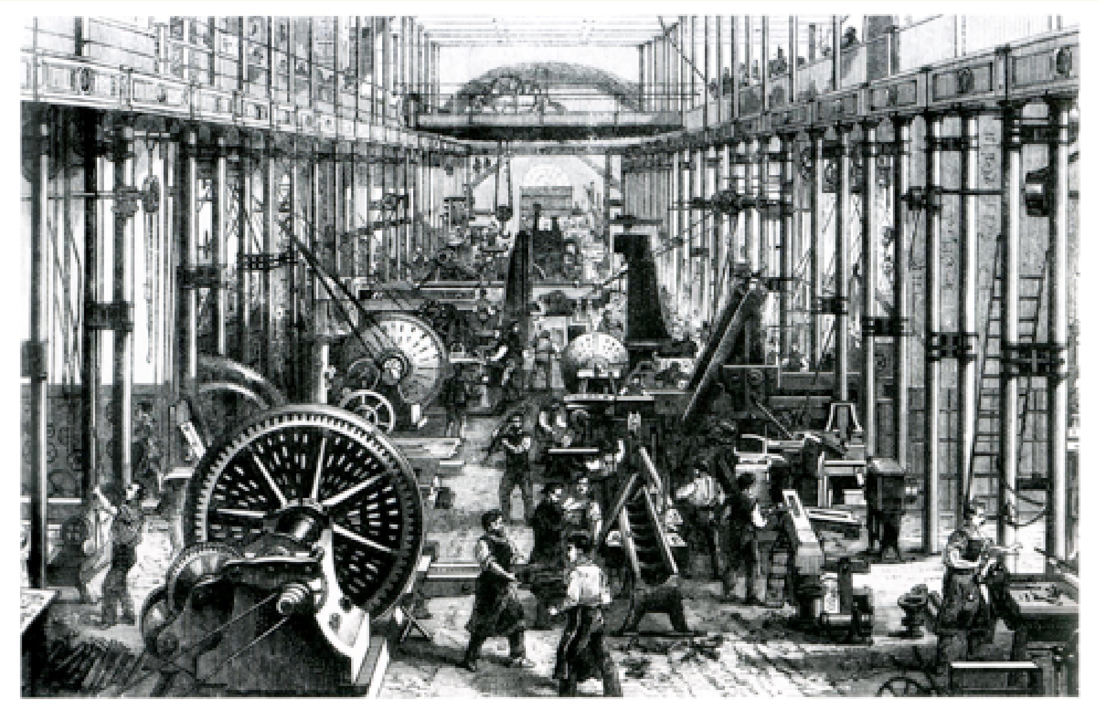


Creative Destruction

- Does technological change destroy jobs or create them?
- The Industrial Revolution brought massive disruption.
- Artisans (like the Luddites) protested the mechanization of their livelihoods.
- Yet, over the 19th and 20th centuries, wages and employment grew rapidly.
- How do we reconcile this? And is today's AI "different"?

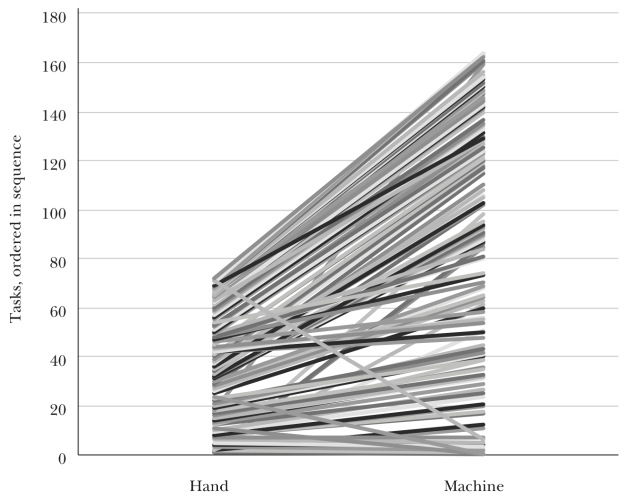
Automation of Manufacturing (1899)

- *Atack, Margo, and Rhode (2019)* use the 1899 "Hand and Machine Labor" study.
- Incredibly detailed observation of how goods were produced before and after factory mechanization.
- Focused on the **task level**: what specifically does a worker do, and how long does it take?
- Example: Making 100 pairs of medium-grade shoes.

The Shift to Extreme Specialization

- Under **hand production**: An artisan performed most tasks.
- The median number of tasks per worker was 2 (often much more for master craftsmen).
- Under **machine production**: The median tasks per worker fell to **1**.
- Workers were intensely specialized.

Figure 1
Slope Chart Linking Hand to Machine Tasks for Unit 71



Source: Authors.
Note: Figure 1 relates each of the various hand tasks for shoe-making on the left to the (far more numerous) machine tasks for shoe-making on the right. Tasks are numbered in sequence. Some hand tasks link to multiple machine tasks. Some are performed in quite different sequences between hand and machine production—these lines cross over. A few hand tasks like "selecting and sorting stock" vanish in machine production (we have connected these to "Task 0" in machine production on the right-hand side). Moreover, the white space on the right-hand axis to which no hand production tasks connect represent new tasks created by mechanization for which there was no hand production analog.

the production of the product (and its supervision). Consequently, any analysis of productivity, including ours, must rely on the measure provided by the study—the amount of time that it took to complete a task—rather than a measure that would be more conventional for economists like value added per worker. Third, while the agents recorded additional information on the survey form that would have been very useful to have for some analyses—for example, the names of the individual workers, and the address of the establishment—this information was not included in the published study. Moreover, as far as we can determine, the completed survey forms have not survived and so this additional information has been lost.[5] Finally,

[5] We tracked down copies of the original survey instrument which are now stored in Record Group (RG) 257 in the US National Archives (US Bureau of Labor Statistics 1890–1905). The forms asked for additional information that was not published, such as the name and location of the establishment, and the names of the workers employed in the production of the various articles.

Productivity Gains from Machine Labor

- On average, machine labor reduced total production time by a **factor of seven**.
- This massive productivity boost came primarily from **steam power**.
- Steam powered the cutting, trimming, and shaping machines that sped up individual tasks.
- But it wasn't just doing the same things faster—it reorganized the work entirely.

Task Transitions

- Mechanization involved subdividing complex tasks and consolidating others.
- Most importantly, it created **new tasks**.
- Fully one-third of tasks in machine production were entirely new!
- Examples: maintaining steam engines, quality inspection, and foreman supervision.

Table 1
Tasks Transitions, Hand to Machine Labor

Transition	*Share of tasks, equal weights*	*Share of tasks, time weights*	*Share using steam power, equal weights*	*Share using water power, equal weights*	*Share using steam power, time weights*	*Share using water power, time weights*
A: Hand Labor						
1:0	0.044	0.030	0.003	0.002	0.002	< 0.001
1:1	0.673	0.604	0.014	0.017	0.009	0.020
1:*M*	0.134	0.192	0.023	0.005	0.008	0.006
N:*M*, *N* > 1, *M* > 1	0.040	0.054	< 0.001	0.010	< 0.001	0.003
N:1	0.108	0.121	0.010	0.018	0.031	0.019
Total	1.000	1.000	0.014	0.014	0.011	0.016
B: Machine Labor						
1:1	0.458	0.563	0.436	0.029	0.461	0.033
1:*M*	0.146	0.172	0.558	0.058	0.538	0.068
N:*M*, *N*>1, *M*>1	0.024	0.038	0.593	0.070	0.518	0.068
N:1	0.037	0.070	0.757	0.051	0.764	0.060
0:1	0.334	0.158	0.360	0.014	0.333	0.020
Total	1.000	1.000	0.444	0.034	0.477	0.040

Source: Computed from a digitized version of the Hand and Machine Labor study, see text and US Department of Labor (1899).
Notes: The unit of observation is a task as described by the staff of the Hand and Machine Labor study. The basic sample size in Panel A is 7,152 hand tasks from 610 production units. The basic sample size in Panel B is 12,473 machine labor tasks from 610 production units. 1:0 transitions are the hand labor tasks that disappeared in the transition to machine labor. 1:1 transitions are hand labor tasks that have a unique counterpart in machine production. In a 1:*M* transition, a single hand labor task subdivides into *M* machine tasks. In an *N*:*M* transition, *N* hand labor tasks transition into *M* machine labor tasks, with *N* > 1 and *M* > 1. In an *N*:1 transition *N* hand tasks combine to a single machine task. 0:1 indicates new tasks under machine labor. For equal weights, observations count equally in determining the average task shares within units. For time weights, observations are weighted by completion time in determining the average task shares within units.

were mechanized, whether by steam or water, similarly weighted. The sample used to compute Table 1 covers 610 of the original 626 manufacturing units.[6] For the most part, our discussion of Table 1 focuses on the equally weighted (rather than the time-weighted) statistics in Table 1.

Although some hand tasks were abandoned in the transition to machine labor, these comprised a small share of hand tasks and of the time spent in hand labor. The largest category of transitions by far was 1:1—that is, the agents were able to match a singleton task in hand production with a singleton task in machine production whose content was deemed to be the same, except that in machine production of the product, the task was far more likely to be mechanized. As can be seen from

[6] We excluded units from foreign countries, those that used horses, and which were otherwise missing data necessary for our analyses.

The Human Capital Impact ("Deskilling")

- Pre-industrial era: "Human capital" meant learning to make a shoe from start to finish.
- Factory era: Workers only needed to learn one specific task on an assembly line.
- This skill could be picked up quickly on the job.
- **Result:** This allowed a rapid, massive expansion of the manufacturing labor force.

Why Did Employment Grow?

- If machines were so efficient, why didn't factories just fire everyone?
- To understand this, we turn to the theoretical framework of *Acemoglu & Restrepo (2019)*.
- The traditional "black box" model of production (where capital and labor are just inputs to a formula) doesn't capture what actually happens.

The Task-Based Model of Production

- Production is modeled as a **continuum of discrete tasks**.
- Some tasks can only be done by humans. Some can be done by capital (machines).
- **Automation** occurs when capital replaces labor in a task.
- This creates a **Displacement Effect**: Labor is pushed out of its traditional roles, decreasing the labor share of value.

Counterbalancing Forces

If automation displaces labor, what saves it?

1. **Productivity Effect:** Machines lower costs, which lowers prices and increases total output. This expands demand for the remaining human tasks.
2. **Reinstatement Effect:** Technology creates entirely **new tasks** where humans have a comparative advantage.

Technology and Labor Demand

We now investigate how technology changes labor demand. We focus on the behavior of the wage bill, *WL*, which captures the total amount employers pay for labor. Recall that

$$\text{Wage bill} = \text{Value added} \times \text{Labor share.}$$

Changes in the wage bill will translate into some combination of changes in employment and wages, and the exact division will be affected by the elasticity of labor supply and labor market imperfections, neither of which we model explicitly in this paper (for discussion, see Acemoglu and Restrepo 2018a, 2018b).

We use this relationship to think about how three classes of technologies impact labor demand: automation, new tasks, and factor-augmenting advances. Consider the introduction of new automation technologies (an increase in *I* in Figure 1). The impact on labor demand can be represented as:

$$\begin{aligned}\text{Effect of automation on labor demand} = {} & \text{Productivity effect} \\ & + \text{Displacement effect.}\end{aligned}$$

The *productivity effect* arises from the fact that automation increases value added, and this raises the demand for labor from non-automated tasks. If nothing else happened, labor demand of the industry would increase at the same rate as value added, and the labor share would remain constant. However, automation also generates a *displacement effect*—it displaces labor from the tasks previously allocated to it—which shifts the task content of production against labor and always reduces the labor share. Automation therefore increases the size of the pie, but labor gets a smaller slice. There is no guarantee that the productivity effect is greater than the displacement effect; some automation technologies can reduce labor demand even as they raise productivity.[5]

Hence, contrary to a common presumption in popular debates, it is not the "brilliant" automation technologies that threaten employment and wages, but "so-so technologies" that generate small productivity improvements. This is because the positive productivity effect of so-so technologies is not sufficient to offset the decline in labor demand due to displacement. To understand when this is likely to be the case, let us first consider where the productivity gains from automation are coming from. These are not a consequence of the fact that capital and labor are becoming more productive in the tasks they are performing, but follow from the ability of firms to use cheaper capital in tasks previously performed by labor. The productivity effect of

[5] Indeed, in Acemoglu and Restrepo (2018b), we show that industrial robots, a leading example of automation technology, are associated with lower labor share and labor demand at the industry level and lower labor demand in local labor markets exposed to this technology. This result is consistent with a powerful displacement effect that has dominated the productivity effect from this class of automation technologies.

Reinstatement: Creating New Jobs

- Automation has always been accompanied by the creation of new roles.
- 19th Century: Engineers, machinists, repairmen, managers.
- 20th Century: Clerical workers, software developers, analysts.
- *Net Impact:* As long as the Productivity and Reinstatement effects outweigh the Displacement effect, labor demand grows!

Evaluating the Modern Era

- Does this historical pattern still hold true today?
- Acemoglu & Restrepo compare two distinct periods:
  - **The Golden Age:** 1947–1987
  - **The Modern Era:** 1987–2017

The Golden Age (1947–1987)

- Rapid automation displaced many workers.
- But it was accompanied by an equally strong **reinstatement effect** (new tasks).
- The net effect on the "task content of production" was zero.
- Wages and employment grew steadily alongside productivity.

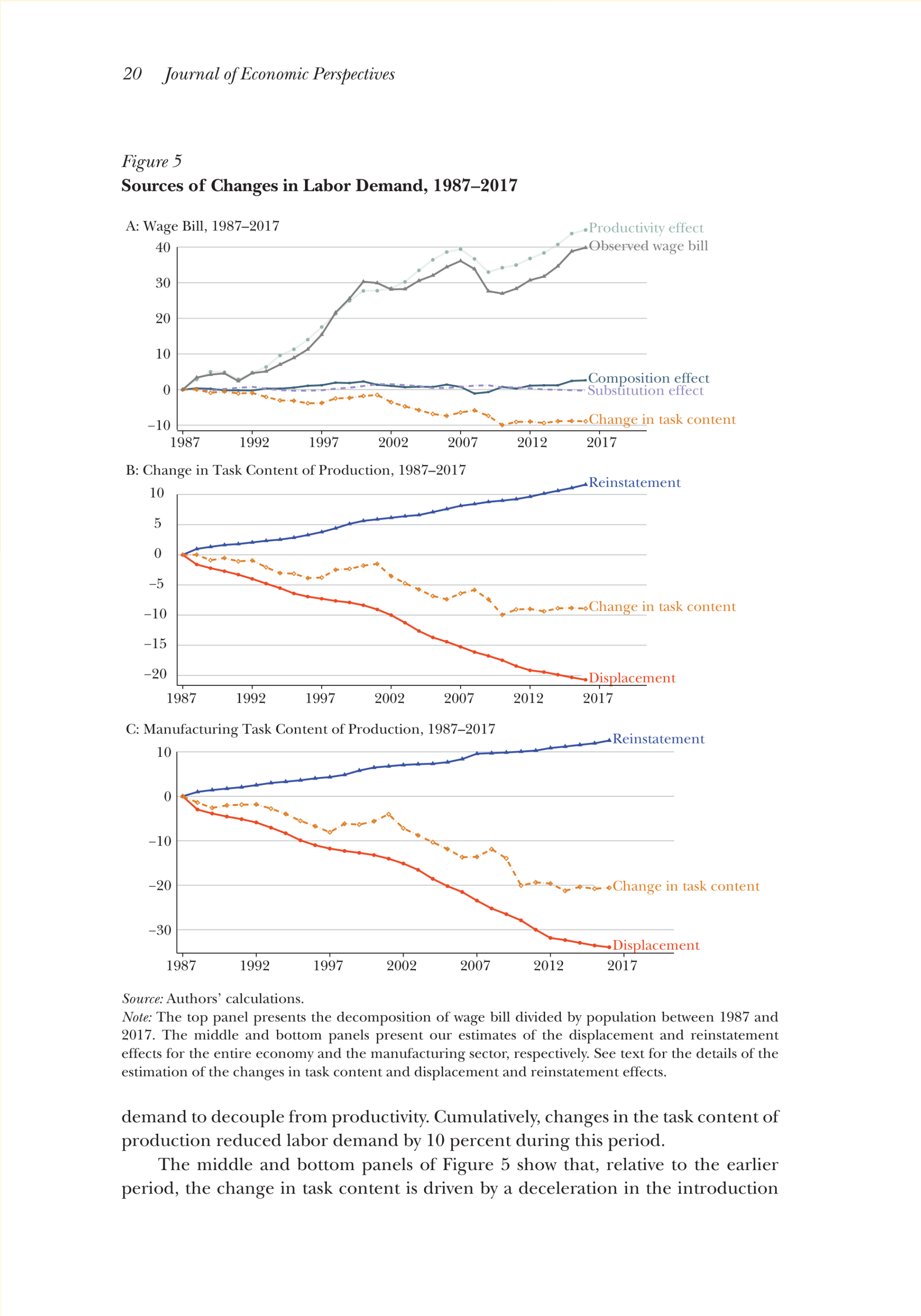

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Figure 5
Sources of Changes in Labor Demand, 1987–2017

Source: Authors' calculations.
Note: The top panel presents the decomposition of wage bill divided by population between 1987 and 2017. The middle and bottom panels present our estimates of the displacement and reinstatement effects for the entire economy and the manufacturing sector, respectively. See text for the details of the estimation of the changes in task content and displacement and reinstatement effects.

demand to decouple from productivity. Cumulatively, changes in the task content of production reduced labor demand by 10 percent during this period.

The middle and bottom panels of Figure 5 show that, relative to the earlier period, the change in task content is driven by a deceleration in the introduction

The Modern Era (1987–2017)

- The story changes significantly in recent decades.
- **Displacement** continued (or accelerated) due to software and robotics.
- But the **Reinstatement Effect** (creation of new tasks for labor) slowed down dramatically.
- The result is a shift *against* labor, causing stagnant wage growth and a falling labor share.

level, and then document the correlation between these measures and our estimates of the change in the task content of production.

We have three measures of industry-level automation technologies. The proxies are: 1) the adjusted penetration of robots measure from Acemoglu and Restrepo (2018b) for 19 industries, which are then mapped to our 61 industries; 2) the share of routine jobs in an industry in 1990, where we define routine jobs in an occupation as in Acemoglu and Autor (2011) and then project these across industries according to the share of the relevant occupation in the employment of the industry in 1990 (see also vom Lehn 2018); and 3) the share of firms (weighted by employment) across 148 detailed manufacturing industries using automation technologies, which include automatic guided vehicles, automatic storage and retrieval systems, sensors on machinery, computer-controlled machinery, programmable controllers, and industrial robots.[11]

Table 1 reports the estimates of the relationship between the change in task content of production between 1987 and 2017 and the proxies for automation technologies and new tasks; each row and column corresponds to a different regression model. The table shows that with all these proxies there is the expected negative relationship between higher levels of automation and our measure of changes in the task content of production in favor of labor (see also visual representations of these relationships in the online Appendix). These negative relationships remain very similar when we add various control variables, including, in column 1, a dummy for the manufacturing sector and, in column 2, imports from China (the growth of final goods imports from China as in Autor, Dorn, and Hanson 2013; Acemoglu, Autor, Dorn, Hanson, and Price 2016) and a measure of offshoring of intermediate goods (Feenstra and Hanson 1999; Wright 2014). Consistent with our conceptual framework, changes in task content are unrelated to imports of final goods from China, but are correlated with offshoring, which often involves the offshoring of labor-intensive tasks (Elsby, Hobijn, and Sahin 2013). Controlling for offshoring does not change the relationship we report in Table 1 because offshoring is affecting a different set of industries than our measures of automation (see the online Appendix).

We also looked at a series of proxies for the introduction of new tasks across industries, and how they are correlated with our measure of the change in task content for 1987–2017. Our four proxies for new tasks are: 1) the 1990 share of employment in occupations with a large fraction of new job titles, according to the 1991 *Dictionary of Occupational Titles* compiled by Lin (2011); 2) the 1990 share of employment in occupations with a large number of "emerging tasks" according to O*NET, which correspond to tasks that workers identify as becoming increasingly

[11] These data are from the Survey of Manufacturing Technologies, and are available in 1988 and 1993 for 148 four-digit SIC industries which are all part of the following three-digit manufacturing sectors: fabricated metal products; nonelectrical machinery, electric and electronic equipment; transportation equipment; and instruments and related products (Doms, Dunne, and Troske 1997). For this exercise, we computed measures for the change in task content of these four-digit manufacturing industries using detailed data from the Bureau of Economic Analysis input-output tables for 1987 to 2007.

Is "This Time" Different?

- Historically, the industrial model of intense specialization paid off.
- Today, being highly specialized in one routine task is a major risk, because AI and robots are increasingly capable of automating specialized roles.
- Modern automation might require a new human capital strategy:
  - A "suite of diverse, relatively uncorrelated skills" as insurance against displacement.
  - A reversal of the 19th-century trend toward extreme specialization.

Summary

- The Industrial Revolution fundamentally reorganized how tasks were allocated to labor and capital.
- While machines displaced artisans, new tasks and rising productivity drove massive labor demand.
- Today's wave of automation may be different if we fail to create new, high-productivity tasks for human workers.